OVERVIEW

Complex problems offer a lot of opportunities for student learning, but how do you keep students moving forward without robbing them of the opportunities to think for themselves? In this conversation, you'll examine how teacher questioning can guide students solving complex problems.

SPARK MEDIA & FOCUS LESSON

Classroom Dialogue
Teacher provides 1:1 support for students working through a complex problem in Big Foot Conspiracy.

PEDAGOGICAL THEME

Fostering Student Discourse

USING THE CONVO GUIDE FOR A GREAT ROUNDTABLE DISCUSSION

- This conversation is designed for a group of 2-5 educators over 30-60 minutes.
- The Convo Guide serves as an agenda for a roundtable conversation where everyone is there both to learn and to share their expertise.
- Each participant should have their own copy of the Convo Guide and access to the Spark Media.
- Pace yourselves to get through the entire agenda in the time you have.
- Questions aren't intended to have a single right answer; instead, they're designed to spark rich discussions.
- At the end, everyone will commit to trying "one small thing" the very next day. Consider how you might follow up with one another!

OBSERVE 5–15 MIN

- To start, take a look at the Spark Media.
- How do the teachers' approaches differ from one another? How would you characterize each approach?
- Which teachers do the best job of moving student thinking forward without taking over the thinking? Why?

DISCUSS 10–15 MIN

- Sometimes our questions to students do not elicit the responses we envision. If a teacher is met with silence or a "wrong" answer, what questions should they ask themselves to decide how to respond?
- In what sort of instances might a teacher not want to respond with a question and instead use a different teacher move?

RELATE 10–15 MIN

- In your own practice, when do you tend to take on more of the burden of thinking than you need to?
- In order to better navigate class discussions so that students own the thinking, what do you need to think about in your planning?

...AND COMMIT! 5–10 MIN

- What is one small thing you can do tomorrow in support of those shifts?

ONE-ON-ON EXCHANGES BETWEEN TEACHERS AND STUDENTS

The following are sample conversations between individual students and Teachers A, B, C, and D. Each relates to the question (below) from the Citizen Math lesson Big Foot Conspiracy in which students apply unit rates to explore pricing conventions.

Q2. For each size, calculate how much a pair of All Stars costs per ounce. Based on this, who would you say is paying the most for shoes and do you think this is fair? Explain.

SIZE	4	6	8	10	12	14	16
PRICE	$50	$50	$50	$50	$50	$50	$50
WEIGHT (OZ)	19.6	22.6	24.2	26	30.2	31.4	37.6
COST PER OUNCE	____	____	____	____	____	____	____

Teacher A

T: What are we trying to find?
S: The price per ounce of each size.

T: What does "cost per ounce" really mean?
S: The price for each ounce of shoe.

T: For size 4, how do we calculate the cost per ounce?
S: ...I'm not sure.

T: For size 4, does each shoe cost $1?
S: No.

T: How do you know?
S: If it did, 19.6 ounces would cost almost $20, not $50.

T: Does each ounce cost $3?
S: No. If it did, the pair would cost $60.

T: Okay. So how much are you paying per ounce? See if you can figure this out for size 4. I'll come back in a few minutes to check in.

Teacher B

T: What are we trying to find?
S: The price per ounce of each size.

T: And how can you do that?
S: I'm not sure.

T: What do you know about the shoes?
S: They're made by Converse. They come in different sizes. Each size costs the same but weighs a different amount.

T: Can you use this to figure out the cost per ounce?

Teacher C

T: What are we being asked to find?
S: The price per ounce of each size.

T: For size 4, what is the price of the pair, in dollars?
S: $50.

T: For size 4, what is the weight of the pair, in ounces?
S: 19.6 ounces.

T: What number do you get when you divide 50 by 19.6?
S: 2.55.

T: And what are the units of that?

T: When you divide dollars by ounces, you get "dollars per ounce."

T: So what are the units of 2.55?
S: Dollars per ounce?

T: Right. So what is the unit rate for size 4?
S: 2.55 dollars per ounce?

T: Good. Now do this for size 16. What do you get?
S: $1.33 per ounce.

Teacher D

T: What are we trying to find?
S: The price per ounce of each size.

T: How do we determine the cost per ounce?
S: I'm not sure.

T: Let's imagine a different scenario. Imagine a 6-pack of soda costs $12. How much would you be paying for each can?
S: $2.

T: How did you get that?
S: I divided $12 by 6 cans.

T: Now imagine that 20 pounds of candy costs $10. How much are you paying for each pound?
S: $10 divided by 20 is 1/2, so $0.50.

T: Now let's return to the shoes. For a size 4, how much does the pair cost and how many ounces does it contain?
S: $50 for 19.6 ounces.

T: So how much are you paying for each ounce?
S: 50 ÷ 19.6 ounces = 2.55, so $2.55 per ounce.

T: How does this compare to size 16?
S: 50 ÷ 37.6 ounces = 1.33, so $1.33 per ounce. The large shoe costs less per ounce than the small shoe.